Project Report

ML Analysis of the Relationship Between Electric Vehicle Charging Infrastructure, House Prices, and Air Quality in the US

Riham Badarna, Mira Bitar

Information Systems Department, University of Haifa, Haifa

1. INTRODUCTION

The rapid adoption of electric vehicles (EVs) has led to substantial investments in charging infrastructure across the United States. While the primary goal of expanding EV charging networks is to support the transition toward cleaner transportation, their presence may also be associated with broader economic and environmental changes within local communities. In particular, the development of charging infrastructure may influence regional property values by increasing accessibility and supporting economic growth, while also contributing to improved air quality by encouraging the adoption of low-emission vehicles.

This project investigates the following research question:

> *Is the presence of EV charging stations associated with higher property values and lower air pollution (NO_2) levels at the county level in the United States from 2023 to 2025?*

To address this question, we integrated multiple publicly available datasets containing information on EV charging stations, housing prices, air pollution, demographic characteristics, and transportation activity. After cleaning and preprocessing the data, we developed and compared two machine learning models: Linear Regression and XGBoost. The models were evaluated using GroupKFold cross-validation and multiple performance metrics, while SHAP analysis and standardized regression coefficients were used to interpret the importance of the selected predictors. The complete source code is publicly available at: `https://github.com/Riham1633/Data-Science`

2. DATA DESCRIPTION AND COLLECTION

2.1. Data Sources

This study combines data from several publicly available sources to investigate the relationship between EV charging infrastructure, property values, and air pollution in the United States. Each dataset provides different information at either the station or county level and was integrated into a single county-year dataset covering the years 2023–2025. Table 1 provides an overview of the datasets used in this study.

The U.S. Department of Energy (DOE) Alternative Fuel Stations dataset was used to obtain information on public EV charging stations. County-level housing prices were collected from the Zillow Home Value Index (ZHVI), while air quality measurements were obtained from the U.S. Environmental Protection Agency (EPA) Air Quality System (AQS). Demographic information, including population and median household income, was obtained from the U.S. Census Bureau, and transportation data were collected from the Federal Highway Administration (FHWA).

Table 1. Datasets used in this study.

Dataset	Source
Alternative Fuel Stations	DOE [1]
Zillow Home Value Index (ZHVI)	Zillow [2]
Air Quality System (AQS)	EPA [3]
American Community Survey (ACS)	U.S. Census Bureau [4]
Highway Statistics	FHWA [5]

2.2. Data Integration

Since the datasets originated from different organizations and were stored in different formats, several preprocessing steps were required before they could be combined. Geographic identifiers were standardized, temporal information was aligned by year, and all datasets were merged into a single county-year dataset covering the years 2023–2025. This integrated dataset was then used throughout the remainder of the analysis.

2.3. Dataset Summary

The final dataset contains 628 county-level observations from 2023 to 2025 and includes 22 variables describing EV charging infrastructure, socioeconomic characteristics, transportation activity, property values, air pollution, and geographic identifiers. The dataset also includes county-level identifiers (e.g., FIPS code, county name, and state) used for data integration and record matching but excluded from model training. Table 2 summarizes the subset of variables used for machine learning, including engineered features created during preprocessing to better represent county-level characteristics and improve model performance.

2.4. Challenges and Limitations

Several challenges were encountered while preparing the data. The datasets were collected by different

Table 2. Variables used for modelling.

Feature	Description	Data Source
log_EV_Count	Log-transformed county-level EV charging station count	DOE
EV_per_100k	EV charging stations per 100,000 residents	Engineered (DOE + Census)
log_DVMT	Log-transformed daily vehicle miles traveled	Engineered (FHWA)
log_Population_Density	Log-transformed population density	Engineered (Census)
median_income	Median household income	Census
Year	Observation year	Extracted from sources
log_Price_Q50	Log-transformed median property value (target)	Zillow
NO2_Q50	Annual county-level median NO_2 concentration (target)	EPA

organizations, resulting in differences in file formats, geographic identifiers, and temporal resolutions. For example, some datasets used ZIP codes while others used county FIPS codes, requiring the use of a ZIP-to-FIPS crosswalk before merging. The data also differed in spatial resolution, ranging from individual EV charging stations and daily air quality measurements to county-level socioeconomic information.

Additional preprocessing was required to create a consistent county-year dataset. This included extracting the observation year from multiple datasets, engineering variables such as population density, and county-level DVMT, and aligning all observations to a common geographic and temporal level.

The resulting integrated dataset served as the foundation for all subsequent preprocessing, feature selection, and machine learning analyses described in the following sections.

3. DATA CLEANING AND PREPROCESSING

Figure 1 summarizes the preprocessing workflow applied to transform the raw datasets into a unified county-level dataset suitable for machine learning. The preprocessing pipeline consisted of data validation, feature engineering, and feature scaling, with each step designed to improve data consistency and model performance.

3.1. Missing Value Strategy

Although the final integrated dataset contained no missing values, the project guidelines required evaluating different imputation strategies. Missing values were therefore simulated by randomly masking observed values, allowing the performance of multiple imputation methods to be assessed under controlled conditions.

Four imputation techniques were evaluated: mean imputation, median imputation, K-Nearest Neighbours (KNN), and Multiple Imputation by Chained Equations (MICE). Each method was tested using two preprocessing orders: imputation followed by logarithmic transformation, and logarithmic transformation followed by imputation. Their performance was evaluated using downstream prediction accuracy, measured by the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). Table 3 summarizes the performance of the evaluated imputation strategies. The Log → Impute pipeline combined with MICE achieved the lowest MAE and RMSE and was therefore selected for the remaining analyses.

Table 3. Comparison of imputation strategies evaluated using downstream prediction performance.

Pipeline	Method	MAE	RMSE
Log → Impute	MICE	**0.2417**	**0.3851**
Log → Impute	KNN	0.4552	0.6044
Impute → Log	KNN	0.7934	1.0598
Log → Impute	Mean	0.9622	1.1952
Impute → Log	Median	0.9516	1.1960
Log → Impute	Median	0.9516	1.1960
Impute → Log	Mean	1.0626	1.3614
Impute → Log	MICE	0.9539	1.4522

3.2. Outlier Detection

Potential outliers were investigated using three complementary approaches: Principal Component Analysis (PCA), Local Outlier Factor (LOF), and Isolation Forest. PCA provided a low-dimensional representation of the dataset, while LOF and Isolation Forest identified observations that differed substantially from their local neighbourhoods or from the overall data distribution.

The detected observations were manually examined to determine whether they represented data-quality issues or naturally occurring county characteristics. Since the identified observations corresponded to valid counties with realistic socioeconomic, transportation, and environmental attributes, no records were removed from the dataset. Retaining these observations preserved the diversity of county-level conditions and avoided unnecessary information loss.

3.3. Feature Engineering

Several feature-engineering steps were performed to harmonize the heterogeneous data sources and derive variables representing county-level EV infrastructure, transportation activity, socioeconomic conditions, property

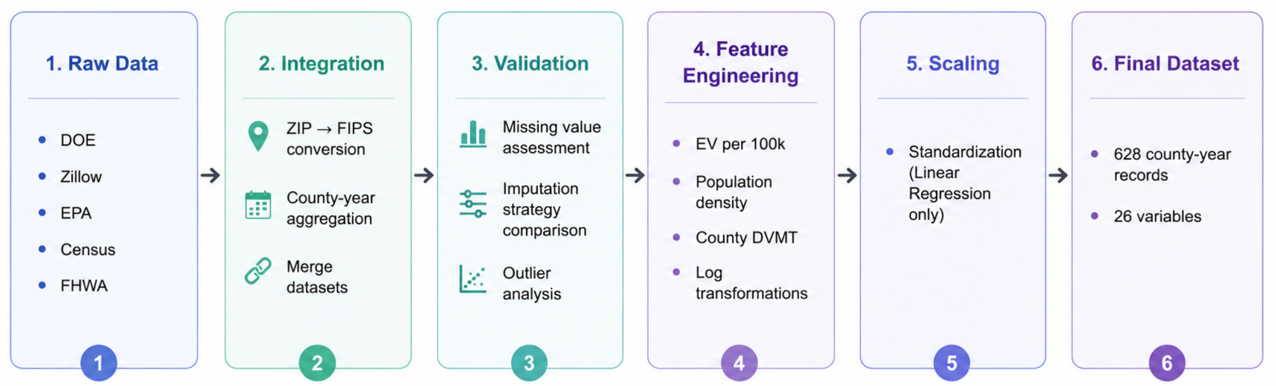


Figure 1. Overview of the data cleaning and preprocessing workflow.

values, and air pollution.

The original datasets differed in both geographic and temporal resolution. Public EV charging stations were reported at the station level, air-quality measurements were available as daily monitoring observations, and the remaining sources used different geographic identifiers. To create a unified county-year dataset, ZIP codes were converted to county Federal Information Processing Standard (FIPS) codes using a ZIP-to-FIPS crosswalk. Station-level EV data were then aggregated into county-year station counts, while daily EPA measurements were aggregated into annual county-level air-quality statistics. The observation year was also extracted from the relevant source files before all datasets were merged.

Transportation activity was represented using Daily Vehicle Miles Traveled (DVMT). The FHWA Highway Performance Monitoring System (HPMS) Spatial All Sections dataset provided the Annual Average Daily Traffic (AADT) and section length for each road segment. Segment-level DVMT was computed as

$$\mathrm{DVMT}_{\mathrm{segment}} = \mathrm{AADT} \times \mathrm{Section\ Length}. \quad (1)$$

County-level DVMT was then obtained by summing the segment-level values across all road sections within the same county. Since complete HPMS data were not available for every study year, the 2024 county-level estimates were used as the baseline. The 2023 and 2025 values were derived by applying the corresponding annual traffic growth rates reported in the FHWA Traffic Volume Trends publications.

County-level median household income was obtained from the U.S. Census Bureau Small Area Income and Poverty Estimates (SAIPE) datasets for 2023 and 2024. Since 2025 county-level income estimates were not available, the most recent available values from 2024 were assigned to the 2025 observations.

Population density was calculated by dividing each county's population by its land area:

$$\mathrm{Population\ Density} = \frac{\mathrm{Population}}{\mathrm{Land\ Area}}. \quad (2)$$

To account for differences in county population size, EV charging infrastructure was also represented as the number of charging stations per 100,000 residents:

$$\mathrm{EV}_{100k} = \frac{\mathrm{EV\ Count}}{\mathrm{Population}} \times 100{,}000. \quad (3)$$

Finally, logarithmic transformations were applied to positively skewed variables, including EV charging-station counts, DVMT, population density, and median property values. These transformations reduced skewness, limited the influence of extremely large counties, and improved the suitability of the variables for regression-based modelling.

3.4. Feature Scaling

Continuous predictor variables were standardized using `StandardScaler` before training the Linear Regression model. Standardization placed the predictors on a common scale, allowing the estimated regression coefficients to be compared more meaningfully. Feature scaling was not applied to XGBoost because tree-based models do not depend on the absolute scale of the predictor variables.

Following data validation, feature engineering, and preprocessing, the final modelling dataset consisted of 628 county-year observations and 29 variables. County identifiers, including the FIPS code, county name, and state, were retained for record matching and data integration but excluded from model training. The resulting dataset was then used for feature selection and model development.

4. MACHINE LEARNING MODELS

Two complementary machine learning models were evaluated to predict county-level median house prices and annual NO2 concentrations using EV infrastructure, transportation, and socioeconomic characteristics. Linear Regression was selected as an interpretable baseline model, while Extreme Gradient Boosting (XGBoost) was chosen to capture potentially nonlinear relationships and feature interactions. Both models were trained using the same set of predictor variables and were evaluated separately for the house price and air pollution prediction tasks.

4.1. Linear Regression

Linear Regression was used as an interpretable baseline model for estimating the relationship between the predictor variables and each target variable. Unlike more complex machine learning algorithms, Linear Regression provides coefficients that directly quantify the expected change in the response variable associated with a one-unit change in each predictor while holding the remaining variables constant. This property makes the model particularly suitable for understanding the relative contribution of different factors to property values and air pollution.

The Linear Regression model is defined as

$$y = \beta_0 + \sum_{j=1}^{p} \beta_j x_j + \varepsilon, \tag{4}$$

where y denotes the target variable, β_0 is the intercept, β_j represents the regression coefficient associated with predictor x_j, p is the number of predictors, and ε is the random error term.

To facilitate coefficient interpretation, all continuous predictor variables were standardized prior to model training.

4.2. XGBoost

Extreme Gradient Boosting (XGBoost) is an ensemble learning algorithm based on gradient-boosted decision trees. Instead of fitting a single model, XGBoost sequentially constructs multiple decision trees, where each tree attempts to correct the prediction errors made by the previous ensemble. This iterative boosting process enables the model to capture complex nonlinear relationships and interactions among predictor variables.

Compared with Linear Regression, XGBoost requires fewer assumptions regarding the underlying relationship between the predictors and the response variable. It naturally models nonlinear effects, automatically captures feature interactions, and is generally robust to multicollinearity and differences in feature scaling. These properties make XGBoost particularly well suited for heterogeneous county-level datasets containing demographic, transportation, environmental, and infrastructure variables.

5. FEATURE SELECTION METHODS

Feature selection was performed to identify an informative subset of predictors while reducing redundancy and multicollinearity. The process combined statistical analyses with model-based feature selection to ensure that the retained variables were both statistically meaningful and beneficial for prediction. Initially, highly correlated variables were identified using Pearson correlation coefficients. Multicollinearity was then evaluated using the Variance Inflation Factor (VIF), followed by Mutual Information (MI) analysis to measure the predictive relevance of each feature. Finally, Recursive Feature Elimination with Cross-Validation (RFECV) was applied using both Linear Regression and XGBoost to determine the optimal feature subset for each prediction task.

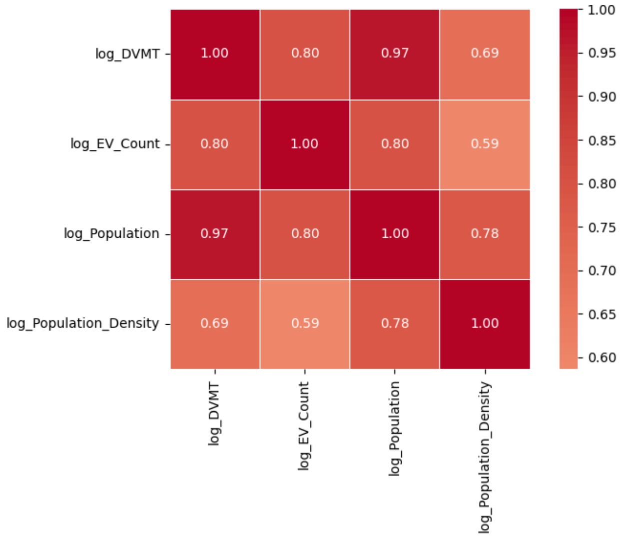


Figure 2. Pearson correlation matrix showing feature pairs with an absolute correlation coefficient of $|r| \geq 0.75$.

5.1. Correlation Analysis

Pearson correlation coefficients were computed for all candidate predictors to identify potentially redundant variables. For visualization purposes, the correlation matrix shown in Figure 2 displays only feature pairs with an absolute Pearson correlation coefficient of $|r| \geq 0.75$, allowing the strongest relationships among the predictors to be highlighted. A more conservative threshold of $|r| \geq 0.80$ was adopted when evaluating whether variables should be removed. Feature pairs exceeding this threshold were further examined based on their interpretability, domain relevance, and potential redundancy before making any elimination decisions.

The correlation analysis revealed strong correlations involving `log_Population`, particularly with `log_DVMT` and `log_Population_Density`. Since population density provides a normalized measure of urbanization while incorporating population information, `log_Population` was excluded from the subsequent analysis.

Although `log_EV_Count` and `log_DVMT` exhibited a correlation close to the removal threshold, both variables were retained because they represent different underlying concepts. Specifically, `log_EV_Count` measures the availability of EV charging infrastructure, whereas `log_DVMT` reflects county-level transportation activity.

Retaining both variables preserves complementary information that is directly relevant to the objectives of this study.

Rather than removing variables solely based on pairwise correlations, highly correlated predictors were further evaluated during the subsequent feature selection stage. This approach allowed statistical evidence to be considered together with model performance before making the final feature selection decisions.

5.2. Variance Inflation Factor

Variance Inflation Factor (VIF) analysis was performed after removing the highly correlated spatial variables to assess multicollinearity among the remaining predictors. A commonly used threshold of VIF < 10 was adopted, indicating that no severe multicollinearity was present. Table 4 presents the VIF values of the retained predictors.

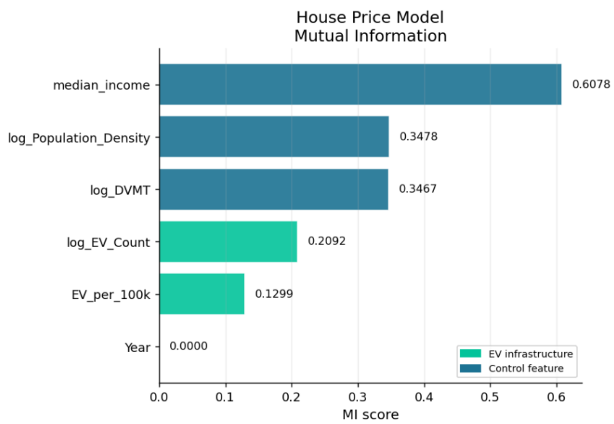
Table 4. Variance Inflation Factor (VIF) values after removing redundant variables.

Feature	VIF
log_EV_Count	6.26
log_DVMT	6.25
EV_per_100k	2.22
log_Population_Density	1.95
median_income	1.26

Although `log_EV_Count` and `log_DVMT` exhibited the largest VIF values, both remained well below the accepted threshold and were therefore retained. Overall, the VIF analysis indicated that multicollinearity was not sufficiently severe to justify removing additional predictors.

5.3. Mutual Information Analysis

Mutual Information (MI) was computed to quantify the amount of information provided by each predictor with respect to the target variables. Unlike correlation analysis, MI captures both linear and nonlinear dependencies, allowing informative variables to be identified even when the underlying relationships are not strictly linear.



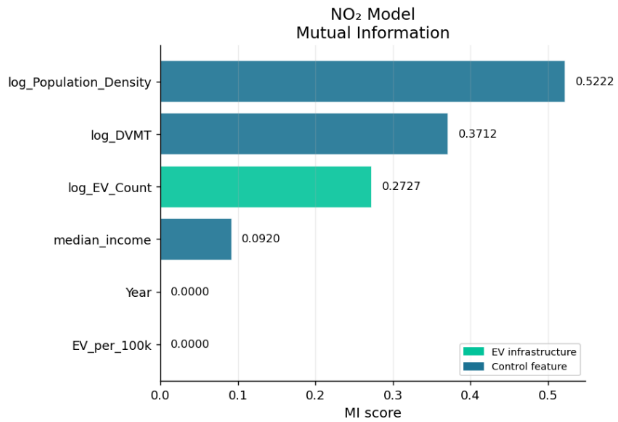


Figure 3. Mutual Information scores for the candidate predictors with respect to the target variable. Higher scores indicate greater relevance for predicting the response.

Figure 3 presents the Mutual Information (MI) scores for all candidate predictors in both prediction tasks. For the House Price model, `median_income` was the most informative predictor, followed by `log_Population_Density` and `log_DVMT`. The EV-related variables (`log_EV_Count` and `EV_per_100k`) exhibited lower MI scores but still contributed useful predictive information.

For the NO_2 model, `log_Population_Density` and `log_DVMT` were the most informative predictors, while `log_EV_Count` showed a moderate association with the target. In contrast, `EV_per_100k` obtained MI scores close to zero, indicating that they provided little information when considered individually. However, these variables were not removed solely based on their MI scores. Instead, the final feature selection decisions were determined using Recursive Feature Elimination with Cross-Validation (RFECV), allowing each feature to be evaluated in the context of the complete predictive model.

Although `Year` exhibited an MI score close to zero, it was retained in the subsequent analysis as a temporal control variable rather than for its individual predictive contribution.

5.4. Recursive Feature Elimination

Recursive Feature Elimination with Cross-Validation (RFECV) was employed to determine the optimal subset of predictors for each prediction task. RFECV recursively removes the least important feature according to the underlying estimator while evaluating model performance using five-fold GroupKFold cross-validation. The feature subset yielding the highest average cross-validation R^2 score is selected automatically.

To ensure that the selected predictors were not dependent on a single modelling approach, RFECV was performed separately using both Linear Regression and XGBoost.

For the house price prediction task, Linear Regression retained all six predictors, while XGBoost selected five predictors, excluding only `Year`. Because `Year` represents the temporal structure of the dataset and was intentionally included to account for changes across the study period, it was retained in the final house price models despite its limited contribution to predictive performance.

For the NO_2 prediction task, both Linear Regression and XGBoost consistently identified `median_income` and `EV_per_100k` as non-essential predictors. Consequently, these two variables were removed from the final NO_2 models. Although both models also excluded `Year`, it was retained because the dataset was partitioned according to the temporal dimension, making `Year` an important control variable for maintaining consistency between training and testing.

The complete feature selection results for both models are provided in Appendix 9.1.

5.5. Final Feature Set

Based on the combined results of correlation analysis, VIF assessment, Mutual Information analysis, and RFECV, different feature sets were adopted for the two prediction tasks.

For the house price model, RFECV retained all six predictors: *log_EV_Count*, *EV_per_100k*, *log_DVMT*, *log_Population_Density*, *median_income*, and *Year*. For the NO_2 model, RFECV consistently indicated that `median_income` and `EV_per_100k` provided little additional predictive value across both Linear Regression and XGBoost. These variables were therefore excluded from the final model, while `Year` was retained as a temporal control variable. The final NO_2 predictor set consisted of *log_EV_Count*, *log_DVMT*, *log_Population_Density*, and *Year*.

This feature selection strategy combines statistical analysis, multicollinearity assessment, information-theoretic measures, and model-based recursive selection to produce compact and interpretable predictor sets while preserving variables that are important for the study design.

6. PERFORMANCE EVALUATION AND CROSS-VALIDATION

6.1. Evaluation Protocol

Model performance was evaluated using a temporal train–test split to assess generalization on unseen data. Models were trained using data from 2023–2024 and evaluated on data from 2025. This setup reflects a realistic prediction scenario by ensuring that information from future observations was not used during training.

To obtain robust estimates of model performance during training, five-fold GroupKFold cross-validation was employed. Counties were used as grouping variables so that observations from the same county were assigned to the same fold, preventing information leakage between training and validation sets. The mean and standard deviation of the cross-validation R^2 scores were used to evaluate both predictive accuracy and model stability.

Linear Regression was trained using standardized predictors, while XGBoost hyperparameters were optimized through GridSearchCV using the same GroupKFold partitions. Model performance was evaluated using the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

6.2. Cross-Validation Performance

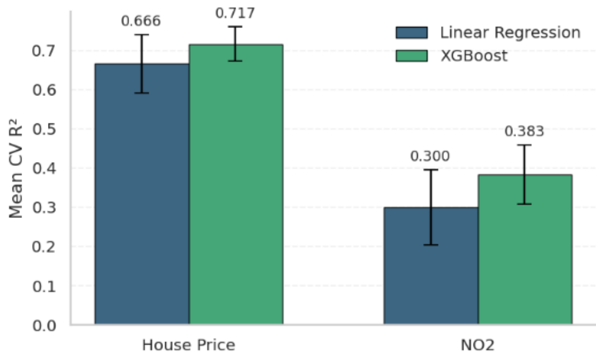


Figure 4. Comparison of five-fold GroupKFold cross-validation performance for Linear Regression and XGBoost. Bars represent the mean cross-validation R^2 across the five folds, while error bars indicate one standard deviation.

Figure 4 compares the mean cross-validation R^2 scores obtained using five-fold GroupKFold validation. Error bars represent one standard deviation across the validation folds.

As expected, XGBoost consistently achieved higher cross-validation performance than Linear Regression for both prediction tasks. In addition, the relatively small standard deviations indicate stable model performance across different county partitions. House price prediction achieved substantially higher R^2 values than NO_2 prediction, suggesting that housing prices exhibit stronger and more predictable relationships with the selected explanatory variables.

6.3. Test Set Performance

Table 5 summarizes the prediction errors obtained on the independent 2025 test set, while Figure 5 compares the corresponding test-set R^2 values.

Table 5. Comparison of Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) for Linear Regression and XGBoost on the independent 2025 test set.

Prediction Task	Model	MAE	RMSE
House Price	Linear Regression	0.207	0.284
House Price	XGBoost	**0.111**	**0.163**
NO_2	Linear Regression	2.116	2.702
NO_2	XGBoost	**1.736**	**2.213**

Figure 5. Comparison of the coefficient of determination (R^2) achieved by Linear Regression and XGBoost on the independent 2025 test set for house price and NO_2 prediction.

The test-set evaluation confirms the findings observed during cross-validation. XGBoost consistently achieved higher predictive accuracy than Linear Regression for both prediction tasks. For house price prediction, XGBoost improved the test-set R^2 from 0.612 to 0.872 while simultaneously reducing both MAE and RMSE by approximately 46% and 43%, respectively. Similar improvements were observed for NO_2 prediction, where XGBoost increased the test-set R^2 from 0.393 to 0.593 and reduced both error metrics. These results suggest that the nonlinear relationships captured by the gradient boosting model provide substantial predictive advantages over the linear baseline.

6.4. Predicted versus Actual Analysis

To further evaluate predictive performance, we compared the predicted and observed values on the test set for both models and prediction tasks. The XGBoost predictions generally lie closer to the identity line than those produced by Linear Regression, indicating improved agreement between predicted and observed values. This behavior is consistent with the quantitative evaluation metrics presented previously, where XGBoost achieved higher R^2 values and lower prediction errors for both house price and NO_2 prediction.

Additional visualizations comparing the predicted and observed values for both models are provided in Appendix 9.2 (Figure 8 and Figure 9), where the XGBoost predictions are shown to follow the identity line more closely than those of Linear Regression.

7. FEATURE ANALYSIS AND IMPORTANCE

7.1. SHAP Feature Importance

To improve the interpretability of the best-performing models, SHAP (SHapley Additive exPlanations) was applied to the fitted XGBoost models. Since XGBoost consistently outperformed Linear Regression on both prediction tasks, SHAP was used to quantify the contribution of each predictor to the model's predictions. Feature importance was computed as the mean absolute SHAP value, where larger values indicate a greater average influence on the predicted outcome.

Figure 6 presents the global SHAP feature importance for the house price prediction model, while Figure 7 shows the corresponding results for the NO_2 prediction model.

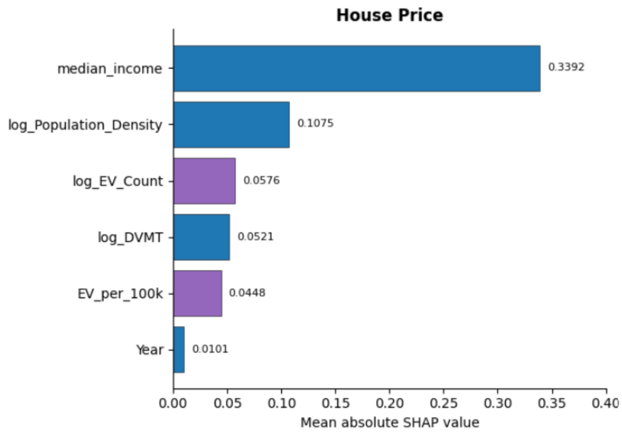


Figure 6. Global SHAP feature importance for the XGBoost house price prediction model. Feature importance is measured as the mean absolute SHAP value.

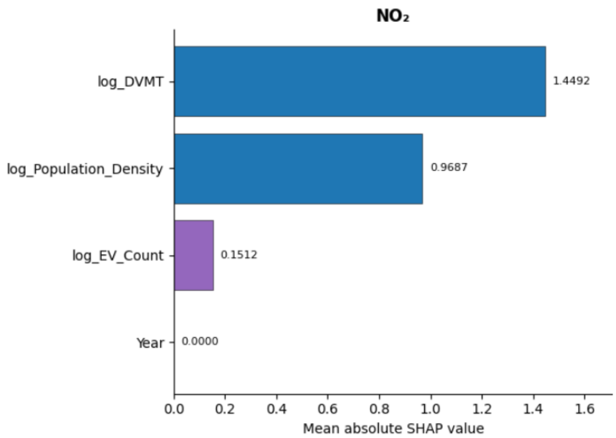


Figure 7. Global SHAP feature importance for the XGBoost NO_2 prediction model. Feature importance is measured as the mean absolute SHAP value.

For house price prediction, *median_income* was identified as the most influential predictor, followed by *log_Population_Density*. The EV-related variables, *log_EV_Count* and *EV_per_100k*, also contributed to the model, although their importance was lower than that of the socioeconomic and demographic variables.

For NO_2 prediction, *log_DVMT* emerged as the dominant predictor, followed by *log_Population_Density*. The contribution of *log_EV_Count* was considerably smaller, while *Year* had a negligible influence on the model predictions.

7.2. Discussion of Feature Importance

The SHAP analysis provides insight into the relative importance of the variables driving each prediction task. For house price prediction, the dominance of *median_income* suggests that local socioeconomic conditions remain the primary determinant of residential property values. Population density also plays an important role, reflecting the influence of urbanization on housing demand and prices. Although EV infrastructure variables were not the strongest predictors, their non-zero SHAP values indicate that they contribute additional predictive information beyond traditional demographic and transportation characteristics.

In contrast, NO_2 concentrations were primarily explained by transportation activity, represented by *log_DVMT*, and by population density. These findings are consistent with the expectation that traffic intensity and urbanization are major contributors to air pollution. The relatively small contribution of *log_EV_Count* suggests that, while EV infrastructure is associated with air quality, its direct influence is modest compared with broader transportation patterns during the study period.

Overall, the SHAP analysis demonstrates that EV infrastructure contributes to both prediction tasks but is not the dominant explanatory factor. Instead, socioeconomic characteristics, urbanization, and transportation activity remain the primary drivers of housing prices and NO_2 levels, with EV infrastructure providing complementary predictive information.

Additional SHAP summary (beeswarm) plots illustrating the distribution and direction of individual feature contributions are provided in Appendix 9.3.

8. RESULTS AND CONCLUSIONS

8.1. Main Findings

This study investigated whether the presence of electric vehicle (EV) charging infrastructure is associated with residential property values and nitrogen dioxide (NO_2) concentrations at the county level in the United States between 2023 and 2025. Two prediction tasks were considered, using both Linear Regression and XGBoost to model house prices and air pollution based on EV infrastructure, transportation activity, demographic characteristics, and socioeconomic variables.

The experimental results demonstrated that XGBoost consistently outperformed Linear Regression for both prediction tasks. The nonlinear model achieved higher predictive accuracy and lower prediction errors, indicating that the relationships between the explanatory variables and the target variables are not purely linear. In particular, XGBoost achieved a test-set R^2 of 0.872 for house price prediction and 0.593 for NO_2 prediction, substantially improving upon the corresponding Linear Regression models.

The SHAP analysis further improved the interpretability of the XGBoost models by identifying the most influential predictors. For house price prediction, median income was the dominant explanatory variable, followed by population density. For NO_2 prediction, transportation activity, represented by log-transformed daily vehicle miles travelled (log_DVMT), and population density were the strongest predictors. Although EV infrastructure variables contributed to both models, their influence was smaller than that of the socioeconomic and transportation-related variables.

Overall, the results suggest that EV charging infrastructure provides complementary predictive information for both housing prices and air quality, while broader demographic, economic, and transportation characteristics remain the primary drivers of these outcomes.

8.2. Limitations and Future Work

Several opportunities exist for extending the present study. First, although the models were evaluated using a temporal train–test split, the analysis covered only three years of data (2023–2025). Incorporating a longer historical time horizon would enable the models to capture long-term trends in EV adoption, housing market dynamics, and environmental conditions.

Second, additional explanatory variables could further improve predictive performance. Examples include weather conditions, industrial emission sources, public transportation availability, land-use characteristics, and charging station capacity or utilization rather than simply the number of charging stations.

Finally, the current analysis treats counties as independent observations. Future work could investigate spatial modelling approaches that explicitly account for geographic relationships between neighbouring counties, such as geographically weighted regression or graph-based machine learning methods. Such approaches may better capture regional patterns in both housing markets and air pollution.

Despite these limitations, the proposed framework demonstrates that combining integrated county-level data, machine learning models, and interpretable AI techniques provides valuable insight into the relationship between EV infrastructure, residential property values, and environmental quality.

References

[1] U.S. Department of Energy, "Alternative Fuels Data Center," [Online]. Available: `https://afdc.energy.gov/`

[2] Zillow Research, "Housing Data," [Online]. Available: `https://www.zillow.com/research/data/`

[3] U.S. Environmental Protection Agency, "Air Quality System," [Online]. Available: `https://www.epa.gov/aqs`

[4] U.S. Census Bureau, "American Community Survey," [Online]. Available: `https://www.census.gov/programs-surveys/acs`

[5] Federal Highway Administration, "Highway Statistics Series," [Online]. Available: `https://www.fhwa.dot.gov/policyinformation/statistics.cfm`

9. APPENDIX

This appendix presents supplementary tables and figures that support the analyses discussed in the main text. The additional material includes the complete RFECV feature selection results, model evaluation visualizations, and SHAP summary (beeswarm) plots.

9.1. RFECV Feature Selection Results

Tables 6 and 7 present the complete Recursive Feature Elimination with Cross-Validation (RFECV) results obtained using Linear Regression and XGBoost for the house price and NO_2 prediction tasks, respectively.

Table 6. RFECV feature selection results for the House Price model.

Feature	LR	XGB
log_EV_Count	✓	✓
EV_per_100k	✓	✓
log_DVMT	✓	✓
log_Population_Density	✓	✓
median_income	✓	✓
Year	✓	–
Optimal # Features	6	5

Table 7. RFECV feature selection results for the NO_2 model.

Feature	LR	XGB
log_EV_Count	–	✓
EV_per_100k	–	–
log_DVMT	✓	✓
log_Population_Density	✓	✓
median_income	–	–
Year	–	–
Optimal # Features	2	3

9.2. Additional Model Evaluation Results

The predicted versus actual plots shown in Fig. 8 and Fig. 9 provide a visual assessment of model performance on the independent test set. Points lying close to the diagonal line indicate accurate predictions, whereas larger deviations represent prediction errors. Consistent with the quantitative evaluation presented in Section 6, the XGBoost models exhibit a closer agreement between predicted and observed values than the Linear Regression models for both house price and NO_2 prediction, demonstrating their superior predictive performance.

Figure 8. Predicted versus actual house price values on the independent 2025 test set for Linear Regression and XGBoost. The dashed line represents perfect agreement between predicted and observed values, while the shaded region corresponds to approximately one Mean Absolute Error (MAE) around the ideal prediction line.

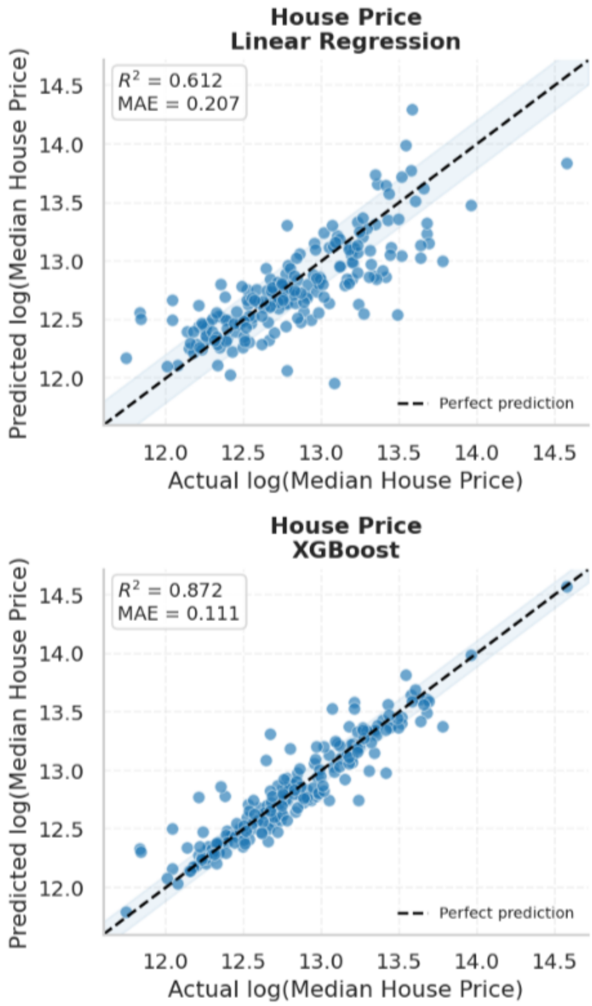
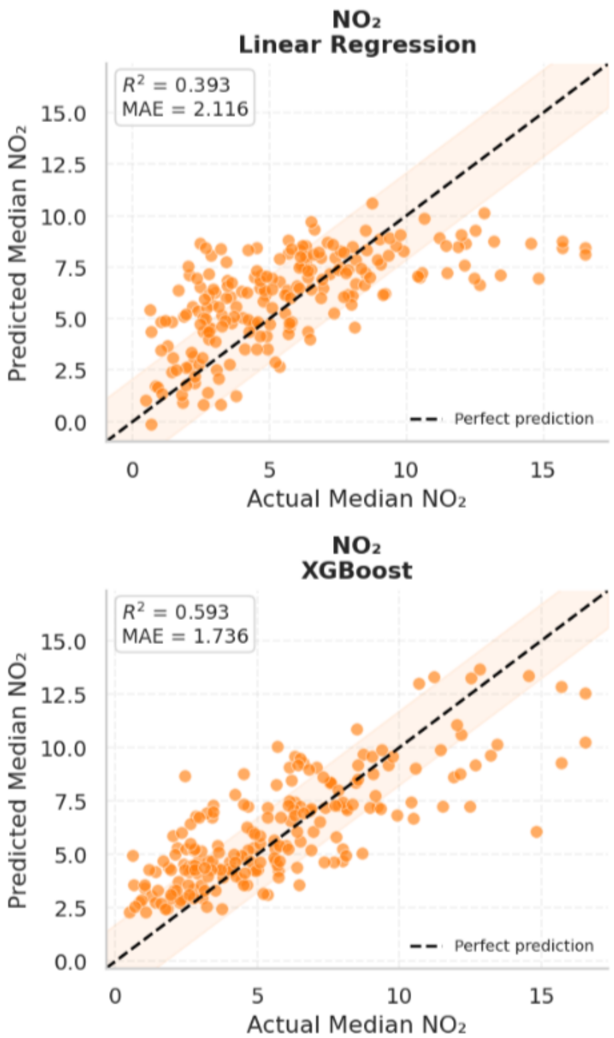


Figure 9. Predicted versus actual NO2 values on the independent 2025 test set for Linear Regression and XGBoost. The dashed line represents perfect agreement between predicted and observed values, while the shaded region corresponds to approximately one Mean Absolute Error (MAE) around the ideal prediction line.



9.3. Additional SHAP Visualizations

The SHAP beeswarm plots presented in this section (Fig. 10 and Fig. 11) provide a more detailed view of feature contributions than the global feature importance plots shown in the main paper. While the importance plots summarize the average magnitude of each feature's contribution, the beeswarm plots additionally illustrate the distribution, direction, and variability of SHAP values across all observations. This enables a more comprehensive interpretation of how individual feature values influence the predictions of the XGBoost models for both house price and NO_2 prediction.

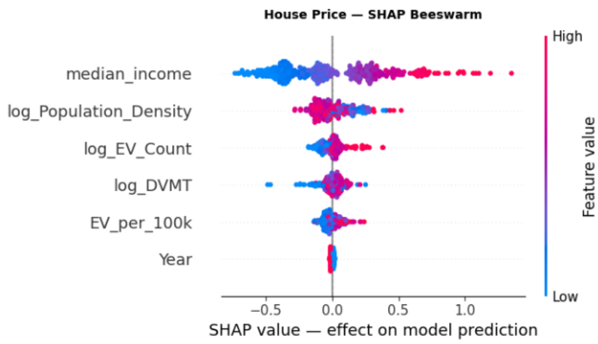


Figure 10. SHAP summary (beeswarm) plot for the XGBoost house price prediction model. The distribution of SHAP values illustrates the contribution of each feature across all observations, while the color indicates the corresponding feature value.

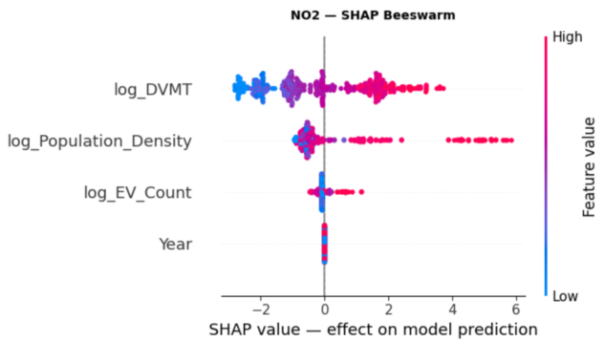


Figure 11. SHAP summary (beeswarm) plot for the XGBoost NO_2 prediction model. The distribution of SHAP values illustrates the contribution of each feature across all observations, while the color indicates the corresponding feature value.